

torch.diag#

<https://pytorch.org/docs/stable/generated/torch.diag.html#torch.diag>

Description:

If input is a vector (1-D tensor), then returns a 2-D square tensor with the elements of input as the diagonal. If input is a matrix (2-D tensor), then returns a 1-D tensor with the diagonal elements of input. The argument `diagonal` controls which diagonal to consider: If `diagonal = 0`, it is the main diagonal. If `diagonal > 0`, it is above the main diagonal.

Function Signature

```
torch.diag(input, diagonal=0, *, out=None) → Tensor#
```

Parameters

Keyword Arguments

Parameters

Keyword

out (Tensor, optional) – the output tensor.

Code Examples

```
>>> a = torch.randn(3)
>>> a
tensor([ 0.5950, -0.0872, 2.3298])
>>> torch.diag(a)
tensor([[ 0.5950, 0.0000, 0.0000],
        [ 0.0000, -0.0872, 0.0000],
        [ 0.0000, 0.0000, 2.3298]])
>>> torch.diag(a, 1)
tensor([[ 0.0000, 0.5950, 0.0000, 0.0000],
        [ 0.0000, 0.0000, -0.0872, 0.0000],
        [ 0.0000, 0.0000, 0.0000, 2.3298],
        [ 0.0000, 0.0000, 0.0000, 0.0000]])
```

```
>>> a = torch.randn(3, 3)
>>> a
tensor([[ -0.4264,  0.0255, -0.1064],
        [  0.8795, -0.2429,  0.1374],
        [  0.1029, -0.6482, -1.6300]])
>>> torch.diag(a, 0)
tensor([ -0.4264, -0.2429, -1.6300])
>>> torch.diag(a, 1)
tensor([ 0.0255, 0.1374])
```

Notes & Warnings

NOTE

See also `torch.diagonal()` always returns the diagonal of its input. `torch.diagflat()` always constructs a tensor with diagonal elements specified by the input.

Detailed Documentation

Documentation

If input is a vector (1-D tensor), then returns a 2-D square tensor with the elements of input as the diagonal.

If input is a matrix (2-D tensor), then returns a 1-D tensor with the diagonal elements of input.

The argument `diagonal` controls which diagonal to consider:

If `diagonal = 0`, it is the main diagonal.

If `diagonal > 0`, it is above the main diagonal.

If `diagonal < 0`, it is below the main diagonal.

`input (Tensor)` – the input tensor.

`diagonal (int, optional)` – the diagonal to consider

`out (Tensor, optional)` – the output tensor.

`torch.diagonal()` always returns the diagonal of its input.

`torch.diagflat()` always constructs a tensor with diagonal elements specified by the input.

Get the square matrix where the input vector is the diagonal:

Get the `k`-th diagonal of a given matrix:

